Z-Score Normalization

Understanding "mean of 0" and "standard deviation of 1"

The words **mean** and **standard deviation** are used in two different roles when we talk about Z-score normalization — once to describe your **original** data, and once to describe a **guaranteed property** that the transformation forces onto your **new** data. A worked example makes this split easy to see.

Step 1 — Start with raw data

Suppose we have five exam scores:

X	50	60	70	80	90

Mean (μ) — the everyday average:

$$\mu = (50 + 60 + 70 + 80 + 90) / 5 = 70$$

Standard deviation (σ) — how spread out the values are around that mean:

$$\sigma = \sqrt{200} \approx 14.14$$

This particular dataset happens to have a mean of 70 and a standard deviation of 14.14. There is nothing special about these two numbers — they are simply whatever the raw data gives us. A different dataset would produce a completely different mean and standard deviation.

Step 2 — Apply the Z-score formula

$$X_{standardized} = (X - \mu) / \sigma$$

Original X	Calculation	Standardized value
50	(50 – 70) / 14.14	–1.414
60	(60 – 70) / 14.14	–0.707
70	(70 – 70) / 14.14	0
80	(80 – 70) / 14.14	0.707
90	(90 – 70) / 14.14	1.414

Step 3 — Check the mean and std of the new numbers

New mean:

$$(-1.414 - 0.707 + 0 + 0.707 + 1.414) / 5 = 0 / 5 = 0$$

New standard deviation: working out the spread of these new values gives exactly **1**.

The Key Distinction

“Mean” and “standard deviation” (no number attached) are just descriptive statistics — they describe where your **original, raw** data happens to sit and how spread out it happens to be. Every dataset has its own different mean and std (70 and 14.14 here — it could be anything for a different dataset).

“Mean of 0” and “standard deviation of 1” are not describing the data anymore — they describe a **guaranteed outcome** that the Z-score formula produces on purpose, every single time, regardless of what the original mean and std were.

• Subtracting μ **forces** the new average to land exactly at 0 (centering the data).
• Dividing by σ **forces** the new spread to shrink or stretch until it measures exactly 1 unit (rescaling the data).

It is not a coincidence, and not something you calculate afterward and hope for — the formula is mathematically built to guarantee this result, no matter what dataset you feed into it.

Why Forcing It to 0 and 1 Matters

This is exactly the normalization problem introduced earlier in the lab. Imagine a second feature, say “income,” originally averaging around 50,000 with a spread of 20,000. If this feature is also standardized, it too ends up with mean = 0 and std = 1 — the exact same “units” as the exam scores above.

Now both features sit on an identical, comparable scale (measured in “how many standard deviations from average”), even though their original raw numbers were wildly different magnitudes (the 70s vs. the 50,000s). That comparability is the entire point — standardization is not just tidying up one feature, it is making **every** feature speak the same numerical language before the network ever sees them.